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Mechanistic understanding of surface instability in polymeric nanofibers under tension

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ABSTRACT

Molecular dynamics and finite element simulations are performed to study the surface instability in polymeric nanofibers under uniaxial tension. A polymeric nanofiber is modeled as a core-shell structure, which has been observed from molecular simulations and experiments. The surface instability of the polymeric nanofiber is found, via finite element simulations and a continuum theory, to be governed by the “polarization” at the core-shell interphase, ultimately being induced by the mismatch between their Poisson’s ratios. Without considering the plastic deformation, the initial imperfections cannot lead to a rippled surface of the polymeric nanofiber under uniaxial tension. It is shown that the strain rate and yield stress greatly influence the onset and modes of the observed rippled surface.